

Question # 1

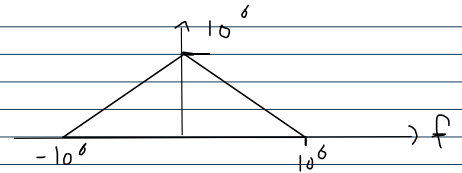
Consider the equivalent filter $p(t) = T(t) * h(t) * R(t)$.

Where $T(t)$ and $R(t)$ are the impulse response of the transmit and receive filter, respectively and $h(t)$ is the impulse response of the channel. If frequency response of $p(t)$ is as follows:

$$P(f) = \begin{cases} 10^6 \left(1 - \frac{|f|}{10^6}\right); & |f| \leq 10^6 \text{ Hz} \\ 0 & \text{otherwise} \end{cases}$$

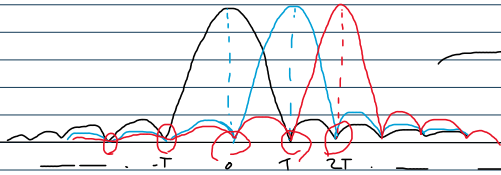
- Draw the frequency response $P(f)$.
- What is the condition for satisfying the Nyquist criteria for ISI free transmission?
- Determine $p(t)$ and verify part (ii).
- Calculate the roll off factor of the filter $p(t)$.

i)



ii) the condition is $\sum_{i=-\infty}^{\infty} s(mT) = \begin{cases} 1; & n=0 \\ 0; & m \neq 0 \end{cases}$

iii) $P(f) = 10^6 \text{tri}\left(\frac{f}{10^6}\right) \xrightarrow{\text{IFT}} P(t) = 10^{12} \text{sinc}^2(10^6 t)$



We can see that at every sampling time the condition $\sum_{i=-\infty}^{\infty} s(mT) = \begin{cases} 1; & m=0 \\ 0; & m \neq 0 \end{cases}$

Only 1 non zero crossing per sample

iv) $\alpha \rightarrow \frac{\text{Excess BW}}{\text{ideal BW}}$

ideal BW $\Rightarrow P(t) = \text{sinc}(10^6 t) \rightarrow \text{BW} = 5 \times 10^5$

Excess BW $= 10^6 - 5 \times 10^5 \rightarrow \alpha = \frac{10^6 - 5 \times 10^5}{5 \times 10^5} = 1 \rightarrow \alpha = 1$

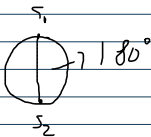
Question # 2

For BPSK with bit rate 0.5 Mbps. Answer the following if the carrier frequency is 1 MHz:

- Draw the transmitted signals representing bit '1' and bit '0' indicating all times and amplitudes. Assume the average bit energy is E_b .
- Show in detail how the impulse response of the matched filter required at the receiver can be determined to recover the transmitted bits.

for bit "1" $\rightarrow 0 < t < T_b$

$$S_1(t) = A_c \cos(2\pi f_c t)$$



for bit "0" $\rightarrow 0 < t < T_b$

$$\theta_1 = 0, \theta_2 = 180$$

$$S_2(t) = A_c \cos(2\pi f_c t + 180)$$

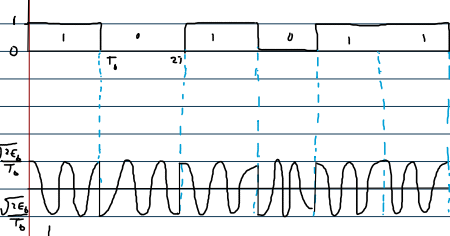
$$= -A_c \cos(2\pi f_c t)$$

using Gram-Schmidt

$$\phi_1(t) = \frac{S_1(t)}{\sqrt{E_b}} \rightarrow \phi_1(t) = \frac{A_c}{\sqrt{E_b}} \cos(2\pi f_c t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t)$$

$$A_c = \sqrt{\frac{2E_b}{T_b}}$$

$$E_b = \int_0^{T_b} S_1(t) dt = \int_0^{T_b} A_c^2 \cos^2(2\pi f_c t) dt = \frac{A_c^2 T_b}{2}$$



ii) matched filter =

$$S(T_b - t)$$

$$y(t) = S(t) * h(t)$$

$$= \int_0^{T_b} \frac{2E_b}{T_b} \cos^2(2\pi f_c t) dt$$

$$= \frac{2E_b}{T_b} \int_0^{T_b} \frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t) dt$$

$$= \frac{2E_b}{T_b} \cdot \frac{T_b}{2} = E_b$$

for bit $\rightarrow -E_b$

if $y(T_b) > 0 \rightarrow 1$

$y(T_b) < 0 \rightarrow 0$